



AgroSonic Defence

IoT-Powered Smart Crop Protection System

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IoT Enabled

ML Powered

Mobile App

Scalable

An intelligent system integrating embedded hardware, real-time sensing, and machine learning to provide automated and adaptive protection for agricultural fields.

12 hrs

Daily Coverage

3 Types

Intruder Classes

ESP32

Core MCU

Real-Time

Notifications

What is AgroSonic Defence?

- IoT-enabled intelligent crop protection system designed to address agricultural intrusion by animals, birds, and humans.
- Integrates PIR and ultrasonic sensors for continuous field monitoring and movement detection.
- ESP32 microcontroller processes sensor data and runs a lightweight ML model to classify the type of intruder.
- Activates targeted deterrents (specific sound frequencies or visual alerts) based on classification results.
- Mobile app integration sends real-time notifications and allows manual farmer control.
- Energy-efficient design operates 12 hours daily during high-risk periods; solar-upgradable.



Embedded Systems

ESP32 MCU with real-time sensor processing



Machine Learning

Lightweight classifier for intruder detection



Mobile Integration

Real-time alerts & manual override controls



Sustainable Design

Low-power, solar-compatible architecture

Agricultural fields face severe crop damage from animal, bird, and unauthorized human intrusions — especially at night — causing significant economic loss and farmer stress.

Night-Time Vulnerability

Intrusions peak after dark when farmers cannot monitor fields, leading to widespread crop destruction and yield loss.

Economic & Social Impact

Repeated crop damage causes financial instability for small-scale farmers and psychological stress affecting overall productivity.

Traditional Methods Fail

Manual guarding, fencing, and basic scare devices are labor-intensive, inconsistent, and ineffective over large areas.

No Intelligent Response

Existing approaches lack automation, classification ability, and real-time alerts — every intrusion is treated the same way.

Existing Solutions & Gaps

Why Current Methods Fall Short

Method	What It Offers	Critical Gap
Manual Guarding	Direct human control	✗ Labor-intensive, unsustainable, risk to farmer safety at night
Physical Fencing	Permanent barrier	✗ Expensive, requires maintenance, ineffective for birds & small animals
Noise Deterrents	Low cost, easy to deploy	✗ Animals habituate over time; no classification or adaptation
Chemical Repellents	Area-wide coverage	✗ Environmental harm, inconsistent effectiveness, costly
Basic Alarm Systems	Automated alert	✗ High false alarms, no intruder classification, no targeted response

Key Gap: No existing solution provides intelligent, automated intruder classification with targeted, adaptive responses and real-time farmer notification.

Proposed Solution

AgroSonic Defence System

AgroSonic Defence combines IoT sensors, ESP32 processing, machine learning classification, and mobile app integration into a fully automated, adaptive crop protection system.

01 Detect



- PIR sensor detects motion
- Ultrasonic sensor measures distance
- Noise filtering validates data
- Dual-sensor confirmation

02 Classify



- Feature extraction from sensor data
- ML model classifies intruder type
- Bird / Animal / Human labels
- Confidence level assigned

03 Respond



- Type-specific sound frequency activated
- LED visual alert triggered
- Mobile push notification sent
- Manual override available

Hardware & Software Architecture

Technical Stack

HARDWARE LAYER

ESP32 MCU

Central Processing Unit

Wi-Fi built-in, low power, fast processing

PIR Sensor

Motion Detection

Detects infrared heat signatures from movement

Ultrasonic Sensor

Distance Measurement

Confirms presence & estimates intruder size

Buzzer Module

Sound Deterrent

Plays frequency-specific deterrent sounds

LED Indicators

Visual Alert

Flashing alerts for visual deterrence

SOFTWARE LAYER

Arduino IDE

Development Environment

Embedded C programming for ESP32

ML Classifier

Intruder Classification

Decision tree / rule-based lightweight model

Wi-Fi Module

Cloud Communication

Sends data to mobile app via HTTP/MQTT

Mobile App

Farmer Interface

Notifications, manual control, activity logs

Model Architecture

Input Features

Motion intensity (PIR) • Distance variation (ultrasonic) • Optional thermal data

Algorithm

Decision Tree / Rule-Based Classifier — compatible with ESP32 limited RAM

Output Classes

 Bird |  Animal |  Human (+ confidence %)

Trigger

Classification fires on validated intrusion only — dual sensor confirmation

Adaptation

Selects specific deterrent sound per class — prevents animal habituation

Advantages

Low latency, no cloud dependency, runs on-device in real time

Classification Flow



Mobile Application & Subscription Model

User Experience & Business Strategy

Mobile App Features

Real-Time Push Alerts

Instant notification with intruder type, time, and detection zone

Manual Override

Farmer can remotely trigger stronger deterrents or disable alerts

System Dashboard

Live sensor status, battery level, Wi-Fi signal strength

Activity Log

Full history of intrusion events, response actions, and timestamps

Alert Customization

Adjust sound intensity, alert schedules, and sensitivity thresholds

Subscription Model

FREE — 6 Months

₹0

- ✓ Full app access
- ✓ Real-time notifications
- ✓ Basic activity logs
- ✓ App support

PRO — Monthly

₹199/mo

- ✓ All Free features
- ✓ Cloud data storage
- ✓ Advanced analytics
- ✓ Priority support + OTA updates

Market Opportunity

Agri-Tech Market Potential



Small & Medium Farmers

Primary target — most affected by crop loss, limited access to advanced technology, price-sensitive buyers.



Agri Cooperatives

Large-scale adoption potential. Can equip thousands of member farms with a centralized subscription model.



Government & NGOs

Subsidized distribution programs for rural farmers aligned with Digital India and PM-Kisan initiatives.

140M+

Farmer Households in India

₹1.5L Cr

Annual Crop Loss Estimate

Smart Farming

Fastest Growing Agri Segment

₹199/mo

Subscription Price Point

The growing adoption of IoT and smart farming solutions, combined with digital literacy among younger farmers, creates a strong opportunity for scalable, subscription-based agri-tech deployment across rural India and beyond.

Technology Stack

Core Technologies Used



Internet of Things (IoT)

Enables seamless communication between sensors, microcontroller, and the mobile application in real time.



Embedded Systems

ESP32 processes sensor data, executes control logic, and manages all hardware outputs through Embedded C on Arduino IDE.



Machine Learning

Lightweight on-device classifier differentiates between birds, animals, and humans from sensor pattern data.



Wireless Communication

Wi-Fi module on ESP32 transmits intrusion events to the cloud and pushes notifications to the farmer's smartphone.



Mobile Application

User-facing app provides real-time alerts, manual control, system status dashboard, and historical activity logs.

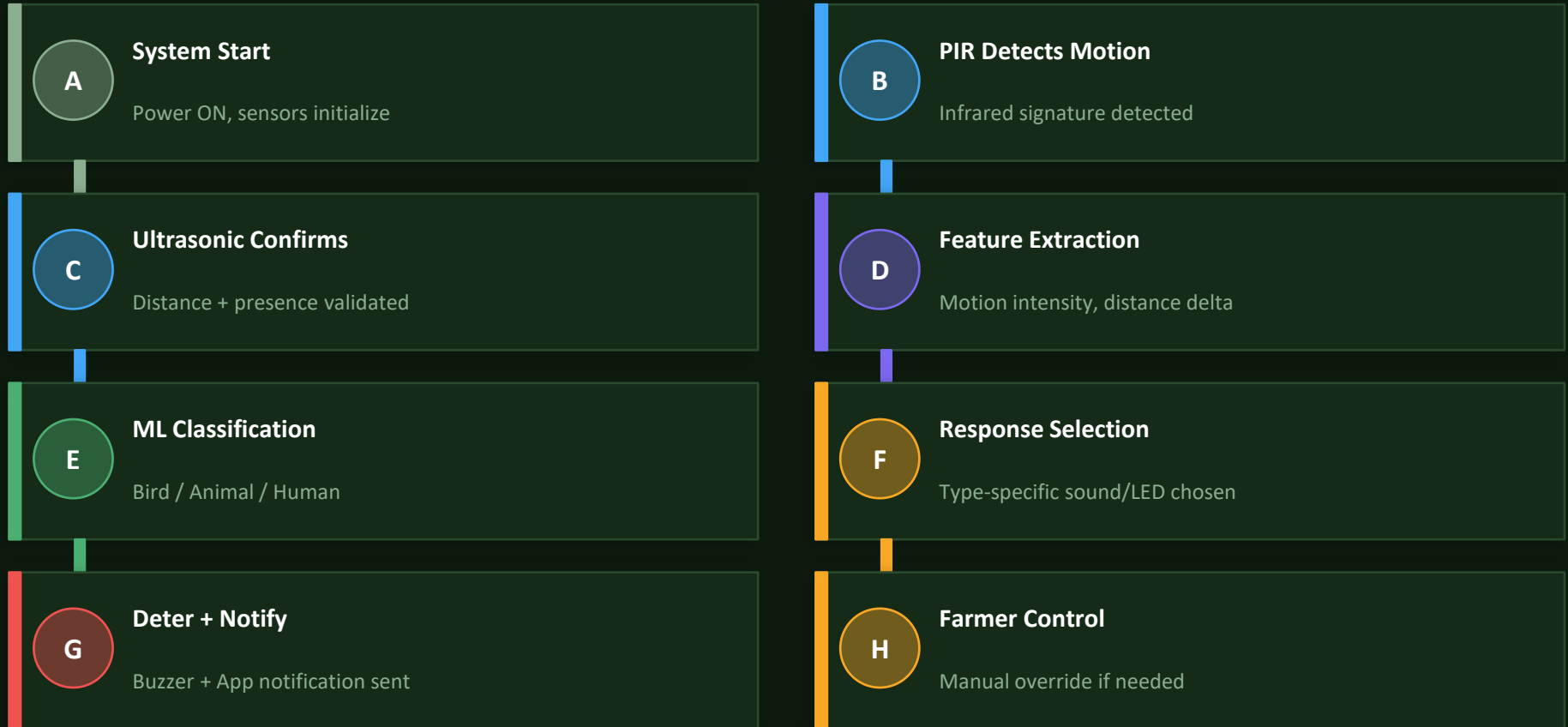


Cloud & Solar (Roadmap)

Future enhancement — cloud analytics for crop protection insights and solar power integration for off-grid deployment.

System Flowchart

End-to-End Process Flow



Feedback Loop: If deterrent fails → Farmer manual override → System logs response → Algorithm improves over time

Project Modules

Functional Architecture

01

Sensing Module

PIR + Ultrasonic sensors continuously monitor the field for motion and intrusion signals

02

Data Acquisition

Collects, filters, and normalizes raw sensor data for downstream processing

03

ML Classification

Lightweight model classifies intruder type with confidence level in real time

04

Decision Module

Determines appropriate deterrent action based on classification output

05

Actuation Module

Drives buzzer with targeted frequency and activates LED visual deterrent

06

Communication

Wi-Fi transmits event data to cloud and mobile app via MQTT/HTTP

07

User Interface

Mobile app provides push alerts, manual control, dashboards, and logs

08

Business Module

Manages subscription lifecycle, service tiers, billing, and software updates

AgroSonic Defence is engineered for practical deployment — low cost, easy installation, and a modular architecture that scales from a single farm to cooperative-level coverage.



Affordability

- Low-cost, off-the-shelf components
- ESP32 < ₹500 on open market
- No specialized hardware required
- Subscription starts free for 6 months



Ease of Use

- No technical expertise needed
- Simple mobile app interface
- Plug-and-play sensor setup
- Farmer can operate independently



Scalability

- Multiple units deployable per farm
- Modular — expand coverage easily
- Cloud-ready for central management
- Supports cooperative-level adoption



Upgradability

- OTA (Over-The-Air) software updates
- Solar power add-on support
- Image recognition upgrade path
- Cloud analytics integration roadmap



AgroSonic Defence

Protecting Farms. Empowering Farmers.

IoT • Machine Learning • Embedded Systems • Smart Agriculture